Unforced Navier–Stokes

Compact symmetric stresses, exact coefficient margins, and formal verification on the three-torus

Tony Newton
Newton Astro Labs, United Kingdom
tony.newton79@gmail.com

Abstract

The absence of external forcing makes the three-dimensional Navier–Stokes regularity problem substantially more restrictive than a construction with prescribed smooth forcing. We study exact mathematical prerequisites for a candidate unforced finite-time singularity on the periodic three-torus. An explicit compact symmetric-divergence inverse is proved for smooth axisymmetric meridional sources supported in an annulus, with vanishing total axial force as the necessary and sufficient compatibility condition. A separate two-diagonal pressure-neutral construction exhibits a sharp radial sign obstruction. Finite seven-ray stress algebra yields the exact coefficient margin 15971/122900 and sharp componentwise amplification factor 14. A compact source-matched dual test excludes a strong physical entrywise stress bound at the corresponding numerical scale. For the frozen cylindrical seven-ray frame, an exact matrix reconstruction identifies that bound with a sufficient four-coordinate H8 stress estimate. We give the exact three-inequality description and sharp dual support function of the original seven-coefficient cone. An exact rational Bernstein certificate shows that even this larger absolute coefficient budget fails for the specified manufactured meridional source. The larger, one-sided seven-ray positivity cone has a distinct asymmetric sharp dual support function. An exact rational enclosure shows that the same dual field does not obstruct this larger cone; no inverse is thereby constructed. A kernel-checked matrix identity then transfers the exact positive-cone dual to any rotating source frame, yielding a necessary integrated bound without assuming the frame is constant. None of these diagnostic tests decides the authentic H8 residual. We derive the exact, jointly sharp hexagonal axial–azimuthal projection of the one-sided cone and a constructive lift to all seven coefficients. The spatial-frame support enclosure retains angular variation of its physical test tensor, repairing an earlier cell-mass approximation. We also derive the exact pressure-compensated annular flux identity and its two sharp frame-dependent cone inequalities. The pressure boundary trace is essential: an axial source primitive alone cannot be bounded by the stress cone when that trace is free. The recovered 122-page Case 006 revision also gives an exact source-native axial interval under the original absolute coefficient budget; its interval is not a necessary condition under one-sided coefficient positivity alone. The source-prescribed separable shell shaper also admits an exact quadratic flux reduction, including its often-leading viscous shear. An endpoint uniqueness lemma rules out an $O(b^2)$ stress-shear budget for any nonzero compact cutoff with a nondegenerate viscous second-jet coefficient, provided the remaining terms have at most first-jet size. The hypothesis is not established for the complete H8 residual. Normalized quarter-turn localization requires twelve ordered nonlinear overlap terms; an explicit local construction shows that these terms can carry three quarters of the core kinetic shear. A separately chosen smooth annular cutoff and normalized toroidal pressure illustrate the remaining source choices, and a combined rational example checks the full flux ledger against direct stress integration. These are test functions, not the original unfrozen H8 cutoff or pressure. For a core localization sharing the shaper cutoff, a cylindrical circulation identity reduces cancellation of the viscous second jet to an exact axial-core moment and one reference-circle trace. For independent core and shaper cutoffs we retain every second spatial derivative, the full axial pressure jump and a rigorous variable-frame enclosure of the H8 shear support. The time derivative of the localized

velocity remains a separate source marginal; its radial primitive cannot be discarded. Lean additionally checks the meridional pointwise inverse and both integral antiderivative identities, alongside the finite cone algebra, physical residual expansion, and conditional terminal-speed retention. These results do not yet construct the source-matched, scale-sharp, positive-cone-compatible compact stress or a genuine zero-force singular solution. The global unforced existence question remains open.

1 The problem and the mathematical objective

We work at viscosity one on the unit three-torus $\mathbb{T}^3 = \mathbb{R}^3/\mathbb{Z}^3$. For a smooth velocity $u(t,x) \in \mathbb{R}^3$ and pressure $p(t,x) \in \mathbb{R}$, the unforced incompressible equations are

$$\partial_t u + (u \cdot \nabla)u - \Delta u + \nabla p = 0, \qquad \operatorname{div} u = 0. \tag{1}$$

The specific target is to find nonzero smooth divergence-free periodic initial data and a solution smooth for $0 \le t < 1$, of finite kinetic energy on a fundamental cell, whose speed becomes unbounded as $t \uparrow 1$. The source-free condition in (1) is essential: a verified construction with a nonzero smooth force does not establish this target. The Clay problem statement makes this distinction explicit [1]. The published OpenAI Lean formalization states Navier–Stokes breakdown with smooth forcing; its unforced finite-time result concerns Euler rather than Navier–Stokes [2].

Our approach isolates small results that are meaningful independently of the unresolved global theorem. First, a source must admit a compact symmetric-divergence correction. Second, if that correction must remain in a prescribed positive stress cone, its components need quantitative control relative to a density envelope. Third, all terms in the actual nonlinear physical residual must cancel with a single periodic pressure. Finally, a singular leading trace can survive only if the correction is provably subleading. Sections 2–7 establish exact parts of this chain.

The construction of compact symmetric anti-divergence operators under force and torque compatibility is not new; see, for example, the support-preserving operator and its explicit moment correction in [3, §3.6 and Appendix A]. The direct axisymmetric formula below is offered as an elementary source-specific coordinate realization, not as a priority claim for compact symmetric anti-divergence. The new obstacle for the present programme is the additional shrinking-support, weighted-seven-ray positivity requirement, which is not supplied by general existence of a compact tensor. Section 5 also shows that a natural *stronger* physical-component envelope is incompatible with one explicit source, even modulo pressure gradients. We subsequently exclude the original absolute coefficient envelope for the same manufactured source by a sharper dual calculation; we then determine the sharp support of the distinct one-sided cone and test the same source against it. The exact frame pullback supplies the correct dual coordinates for the genuine spatially varying source. The actual H8 source remains a separate problem.

2 A compact symmetric inverse in an annular region

Let $0 < a < b$ and $c < d$, and write $\Omega = (a,b) \times (c,d)$ in meridional cylindrical coordinates (r,z). All functions in this section are independent of the azimuthal angle θ. The orthonormal cylindrical frame is (e_r, e_θ, e_z). For a symmetric tensor with components

$$S = K\, e_r \otimes e_r + L\, e_\theta \otimes e_\theta + M(e_r \otimes e_z + e_z \otimes e_r) + C\, e_z \otimes e_z, \tag{2}$$

standard cylindrical differentiation gives

$$(\operatorname{div} S)_r = \partial_r K + \frac{K-L}{r} + \partial_z M, \tag{3}$$

$$(\operatorname{div} S)_z = \partial_r M + \frac{M}{r} + \partial_z C, \tag{4}$$

$$(\operatorname{div} S)_\theta = 0.$$

The $1/r$ terms are geometric, not lower-order terms that may be discarded near a shrinking cylindrical support.

Theorem 2.1 (Explicit compact meridional inversion)**.** *Let $f_r, f_z \in C_c^\infty(\Omega)$, and extend $f = f_r e_r + f_z e_z$ by zero to $\mathbb{R}^3$ in cylindrical coordinates. There is a symmetric $S \in C_c^\infty(\mathbb{R}^3; \mathrm{Sym}_3)$, axisymmetric and supported in the same open annular cylinder, with $\operatorname{div} S = f$, if and only if*

$$\int_a^b \int_c^d r f_z(r,z)\,\mathrm{d}z\,\mathrm{d}r = 0. \tag{5}$$

The construction uses no nonlocal spatial projection, but it asserts no positive-semidefinite or weighted-cone bound on S.

Proof. For necessity, integrate the axial component of $\operatorname{div} S = f$ against the cylindrical Jacobian r. Compact support and (4) give

$$\int_a^b \int_c^d r f_z\,\mathrm{d}z\,\mathrm{d}r = \int_a^b \int_c^d \left[\partial_r(rM) + r\partial_z C\right]\mathrm{d}z\,\mathrm{d}r = 0.$$

For sufficiency, define the axial source marginal and its radial primitive by

$$J(r) = \int_c^d f_z(r,z)\,\mathrm{d}z, \qquad H(r) = \frac{1}{r}\int_a^r sJ(s)\,\mathrm{d}s. \tag{6}$$

Since $r > 0$ on $[a,b]$, H is smooth. The vanishing moment (5) makes H identically zero outside a compact subinterval of (a,b). Choose any $\eta \in C_c^\infty((c,d))$ normalized by $\int_c^d \eta(z)\,\mathrm{d}z = 1$, and set

$$M(r,z) = H(r)\eta(z). \tag{7}$$

The identity $(rH)' = rJ$ implies

$$\int_c^d \left(f_z - \partial_r M - \frac{M}{r}\right)\mathrm{d}z = J - H' - \frac{H}{r} = 0.$$

Consequently the axial primitive

$$C(r,z) = \int_c^z \left[f_z(r,\zeta) - \partial_r M(r,\zeta) - \frac{M(r,\zeta)}{r}\right]\mathrm{d}\zeta \tag{8}$$

vanishes both below and above the axial support; it is smooth and compactly supported in Ω. Choose any $K \in C_c^\infty(\Omega)$, including $K = 0$, and define

$$L = K + r\big(\partial_r K + \partial_z M - f_r\big). \tag{9}$$

Each component of (2) is smooth and compactly supported away from the axis. Substitution in (3) gives

$$\partial_r K + \frac{K-L}{r} + \partial_z M = \partial_r K - (\partial_r K + \partial_z M - f_r) + \partial_z M = f_r,$$

and (8) gives $\partial_r M + M/r + \partial_z C = f_z$. Thus $\operatorname{div} S = f$ as claimed. □

Remark 2.2 (Scope of the construction)**.** The theorem concerns a meridional source with $f_\theta = 0$ and support strictly separated from $r = 0$. An azimuthal source introduces a different angular-momentum compatibility condition, and extending the construction across the axis requires separate regularity conditions on the cylindrical components. The tensor in the theorem may be indefinite, and its size can deteriorate when the support shrinks. None of these omissions is harmless for the weighted H8 attachment problem.

3 Pressure-neutral gauges and a sign obstruction

The mixed and hoop degrees of freedom in Theorem 2.1 are not optional decorations. A restricted diagonal ansatz exhibits a genuine obstruction to maintaining one sign.

Proposition 3.1 (Two-diagonal gauge)**.** *Let $F \in C_c^\infty(\Omega)$, put $K = rF_r$ and $C = F + rF_r$, and take $S = Ke_r \otimes e_r + Ce_z \otimes e_z$. Then*

$$\operatorname{div} S = \nabla C. \tag{10}$$

In particular its divergence is a pressure gradient, so it cannot by itself represent a prescribed source with a nonzero solenoidal part.

Proof. The radial divergence is $K_r + K/r$. Since $K = rF_r$, $C_r = F_r + K_r = K/r + K_r$. The axial divergence is C_z, which proves (10). □

Proposition 3.2 (Restricted radial sign condition)**.** *If K and C are compactly supported in $r \in (a, b)$ at a fixed z and obey $C_r = K_r + K/r$, then*

$$\int_a^b K(r, z)/r \, \mathrm{d}r = 0. \tag{11}$$

Consequently, $K \geq 0$ forces $K \equiv 0$ for this restricted two-diagonal class.

Proof. Integrate the derivative identity between a and b; the endpoint terms of K and C vanish by compact support. Since $r > 0$, continuity and (11) imply that a nonnegative K is identically zero. □

This conclusion does not exclude a sign-changing correction around a positive background tensor or a compact construction with nonzero hoop and mixed components. Indeed, Theorem 2.1 explicitly uses both of those components. The more difficult question is whether one can choose them while retaining a prescribed strict coefficient cone with uniform shrinking-support estimates.

4 The exact seven-ray coefficient margin

The finite cone appearing in the current H8 source has seven rays. With aperture $t = 1/4$ they are

$$\begin{aligned} &v_1 = (1, 0, 0), \quad v_2 = (1, t, 0), \quad v_3 = (1, -t, 0), \\ &v_4 = (1, 0, t), \quad v_5 = (1, 0, -t), \\ &v_6 = (1, t, t), \quad v_7 = (1, t, -t). \end{aligned} \tag{12}$$

Their normalized positive coefficient centre is $c_i^{(0)} = 236/1229$ for $i = 3$ and $172/1229$ for every other index. The raw source covariance and its trace-free rank-one identity are established by exact finite arithmetic and in `UnforcedPatching.SevenRayCenter.lean`.

Write $(A, B, C, D) \in \mathbb{R}^4$ for the four trace-free stress coordinates. The resulting coefficient perturbations, in ray order, are

$$\delta c = \left(-\tfrac{3}{2}D,\ 2C - 4D,\ -2C - 4D,\ 2A - 8B + 4D,\ -2A + 8B + 4D,\ 8B,\ -8B\right). \tag{13}$$

Proposition 4.1 (Sharp componentwise cone estimate)**.** *For $m = \max(|A|, |B|, |C|, |D|)$ the map (13) satisfies*

$$\max_{1 \le i \le 7} |\delta c_i| \le 14m, \tag{14}$$

and the number 14 is optimal. If $\rho \ge 0$ and

$$m \le \frac{15971}{1720600}\rho, \tag{15}$$

then for every ray $c_i^{(0)}\rho + \delta c_i \ge \rho/100$.

Proof. The absolute row sums in (13) are respectively 3/2, 6, 6, 14, 14, 8, 8; hence the operator norm from $\ell^\infty(\mathbb{R}^4)$ to $\ell^\infty(\mathbb{R}^7)$ is at most 14. At $(A, B, C, D) = (1, -1, 0, 1)$, the fourth coordinate equals 14, so the constant is sharp. Since

$$\frac{172}{1229} - \frac{1}{100} = \frac{15971}{122900},$$

inequality (15) implies $|\delta c_i| \le 14m \le (15971/122900)\rho$. As $c_i^{(0)} \ge 172/1229$, the claimed floor follows. □

The seven-ray claim is a finite linear-algebra statement. It provides the exact smallness target for a potential compact correction but gives no estimate for the particular tensor delivered by Theorem 2.1. In particular, no implication from unweighted compact solvability to (15) is available.

Proposition 4.2 (The frozen-frame physical tensor map)**.** *In the local co-rotating frame $(e_1, e_2, e_3) = (e_r, e_\theta, e_z)$ of the seven-ray construction, let $G = \sum_{i=1}^7 \delta c_i v_i \otimes v_i$. The exact physical tensor perturbation and its trace-free pressure representative are*

$$G = \begin{pmatrix} -\frac{3}{2}D & C & A \\ C & -\frac{1}{2}D & B \\ A & B & \frac{1}{2}D \end{pmatrix}, \quad H = G + \frac{D}{2}I = \begin{pmatrix} -D & C & A \\ C & 0 & B \\ A & B & D \end{pmatrix}. \tag{16}$$

In particular, if $m = \max(|A|, |B|, |C|, |D|)$, then $\max_{i,j} |H_{ij}| = m$ in that frame. Conversely, the coordinates are recovered as $A = H_{rz}$, $B = H_{\theta z}$, $C = H_{r\theta}$, and $D = H_{zz}$. The image is the four-dimensional subspace of symmetric trace-free matrices with $H_{\theta\theta} = 0$ (hence $H_{rr} = -H_{zz}$); one independent trace-free diagonal mode is absent. For a different orthonormal frame with columns E, the physical matrix is EHE^{T}; its componentwise maximum need not equal m.

Proof. Substitute the seven vectors (12) and their coefficient perturbations (13) into the six independent entries of G. The diagonal entries are $-3D/2, -D/2, D/2$; the off-diagonal entries in $(r\theta, rz, \theta z)$ order are (C, A, B). Thus $\operatorname{tr} G = -3D/2$, and the scalar gauge addition $DI/2$ removes its trace. The maximum-entry identity then follows directly from the displayed matrix. Conjugation gives the last claim. □

Proposition 4.3 (The exact original seven-coefficient cone)**.** *For any $\tau \ge 0$, all seven constraints $|\delta c_i| \le \tau$ of (13) hold if and only if*

$$4|D| + 2|C| \le \tau, \qquad 4|D| + 2|A - 4B| \le \tau, \qquad 8|B| \le \tau. \tag{17}$$

The sufficient bound $m \le \tau/14$ therefore need not hold for a valid coefficient perturbation: at $(A, B, C, D) = (\tau, \tau/8, 0, 0)$, the exact largest coefficient perturbation and m both equal τ.

Proof. For real x, y the identity $\max(|x+y|, |x-y|) = |x| + |y|$ is exact. Apply it to the paired coefficients $(\delta c_2, \delta c_3)$, $(\delta c_4, \delta c_5)$, and $(\delta c_6, \delta c_7)$. The first pair contributes $4|D| + 2|C|$, the second $4|D| + 2|A - 4B|$, and the third $8|B|$. The remaining coefficient $|\delta c_1| = 3|D|/2$ is dominated by the first pair. Substitution verifies the equality case. □

This identifies a previously implicit assumption: the bound (15) is also a physical entrywise bound for the trace-free perturbation *in the specified cylindrical frame*. The same conclusion requires a verified change of frame when H8 rays are tilted; pointwise orthogonal conjugation alone does not control the additional derivatives of a spatially varying frame.

5 A source-matched weighted dual obstruction

The coefficient condition (15) involves four trace-free coordinates, not arbitrary physical tensor components. To test a particularly strong candidate estimate, let

$$P(r) = \begin{cases} (r-1)^2(2-r)^2, & 1 \le r \le 2, \\ 0, & \text{otherwise}, \end{cases} \qquad \eta(z) = \begin{cases} (1-z^2)^3, & |z| \le 1, \\ 0, & \text{otherwise}. \end{cases} \tag{18}$$

Set $\rho(r, z) = P(r)\eta(z) \ge 0$ and choose the meridional source $f = P(r)\eta'(z)e_z$. This is source-matched in the literal sense that $S = \rho\, e_z \otimes e_z$ is a compact symmetric inverse: $\operatorname{div} S = f$. Also $\int r f_z \,\mathrm{d}r\,\mathrm{d}z = 0$, so the force compatibility condition of Theorem 2.1 holds.

Proposition 5.1 (A pressure-independent stress bound)**.** *Let $\beta = 15971/1720600$. No smooth compact symmetric tensor H and smooth scalar pressure q satisfy both*

$$\begin{aligned} \operatorname{div} H + \nabla q &= f, \\ |H_{ij}| &\le \beta\rho \quad (i, j \in \{r, \theta, z\}), \end{aligned} \tag{19}$$

where components refer to the orthonormal cylindrical frame*. The contradiction also persists for suitable smooth compact nonnegative approximations to the profiles in* (18)*.*

Proof. Construct an axisymmetric vector field with components

$$v_r = -rP\eta'', \qquad v_\theta = 0, \qquad v_z = (2P + rP')\eta'. \tag{20}$$

It is divergence-free because $r^{-1}\partial_r(rv_r) + \partial_z v_z = 0$. The zero-extended profiles give a compactly supported Lipschitz v: $P = P' = 0$ at $r = 1, 2$, and $\eta' = \eta'' = 0$ at $z = \pm 1$. Thus integration by parts against v is legitimate, by approximation if necessary, and the pressure term vanishes. With the angular factor 2π suppressed, the source pairing is

$$\begin{aligned} L &= \int_1^2 \int_{-1}^1 r f_z v_z \,\mathrm{d}z\,\mathrm{d}r \\ &= \left(\int_1^2 rP^2 \,\mathrm{d}r\right)\left(\int_{-1}^1 (\eta')^2 \,\mathrm{d}z\right) = \frac{1}{420}\,\frac{1024}{385} = \frac{256}{40425}. \end{aligned} \tag{21}$$

Here the middle equality follows by integrating $r^2 PP'$ by parts. The physical cylindrical gradient of v includes its hoop entry v_r/r. Its entrywise absolute sum is bounded by

$$|\nabla v|_1 \le (4P + 2r|P'|)|\eta''| + rP|\eta'''| + (3|P'| + r|P''|)|\eta'|. \tag{22}$$

The elementary exact radial integrals are $\int_1^2 P = 1/30$, $\int_1^2 P^2 = 1/630$, $\int_1^2 rP^2 = 1/420$, and $\int_1^2 P|P'| = 1/256$. On their supports, $r \le 2$, $|P''| \le 2$, $|\eta''| \le 6$ and $|\eta'''| \le 72|z| + 120|z|^3$. Direct integration gives

$$\int_{-1}^1 \eta = \frac{32}{35}, \quad \int_{-1}^1 \eta|\eta'| = 1, \quad \int_{-1}^1 \eta|z| = \frac{1}{4}, \quad \int_{-1}^1 \eta|z|^3 = \frac{1}{20}.$$

Consequently, writing $U = \int_1^2 \int_{-1}^1 r\rho|\nabla v|_1 \,\mathrm{d}z\,\mathrm{d}r$, one obtains the rigorous upper bound

$$U \le \left(\frac{4}{420} + \frac{2}{64}\right)\frac{192}{35} + \frac{2}{315}\,24 + \frac{3}{128} + \frac{4}{15} = \frac{313361}{470400}. \tag{23}$$

But (19) would imply $L \le \beta U$ by the virtual-work identity $\int f \cdot v = -\int H : \nabla v$, whereas

$$L - \beta\,\frac{313361}{470400} = \frac{443015653}{2967690880000} > 0, \tag{24}$$

a contradiction. Nonnegative convolution approximations of P and η, supported away from the axis, converge in the requisite derivative integral norms. The strict gap survives for sufficiently small smoothing parameters and proves the smooth-profile assertion. □

Corollary 5.2 (Failure of the sufficient four-field envelope)**.** *For the source f and envelope ρ in (18), there is no fixed-frame four-coordinate stress $H(A,B,C,D)$ of (16) satisfying $\operatorname{div} H + \nabla q = f$ and the sufficient bound (15), even if the pressure is freely chosen.*

Proof. Proposition 4.2 transfers (15) to the entrywise physical cap (19). This contradicts Proposition 5.1. □

The preceding argument excludes the *stronger sufficient* four-field estimate. It cannot alone exclude the original absolute coefficient budget $\max_i |\delta c_i| \le (15971/122900)\rho$: if one simply inserts that larger constant into the conservative gradient bound (23), then

$$\frac{15971}{122900}\frac{313361}{470400} - L = \frac{51024386641}{635933760000} > 0. \tag{25}$$

The sign of this bound is inconclusive, not evidence of existence. The exact cone, however, has a smaller sharp dual support than that componentwise comparison.

Proposition 5.3 (Sharp dual of the original absolute coefficient cone)**.** *For $\tau \ge 0$ let $\mathcal{K}_\tau$ be the four-coordinate set defined by (17). For arbitrary $a,b,c,d \in \mathbb{R}$,*

$$\sup_{(A,B,C,D)\in\mathcal{K}_\tau} (aA + bB + cC + dD) = \tau\left[\max\left(\frac{|a|+|c|}{2}, \frac{|d|}{4}\right) + \frac{|4a+b|}{8}\right]. \tag{26}$$

The upper bound in (26) is proved in Lean for the actual seven coefficient constraints.

Proof. Put $x = A - 4B$. By (17), for fixed $s = |D| \in [0, \tau/4]$, the variables x, C, B are independent and satisfy

$$|x|, |C| \le \frac{\tau - 4s}{2}, \qquad |B| \le \frac{\tau}{8}.$$

The objective becomes $ax + (4a+b)B + cC + dD$. Its maximum for this s is

$$\frac{\tau - 4s}{2}(|a| + |c|) + \frac{\tau}{8}|4a+b| + s|d|.$$

This expression is affine in s, so its maximum occurs at $s = 0$ or $s = \tau/4$, yielding (26). Each bound is attained by choosing the signs of x, C, B, D to match their respective linear coefficients. Thus the value is sharp. □

Proposition 5.4 (An exact original-budget obstruction for the manufactured source)**.** *Let f, ρ be given by (18), and set $\gamma = 15971/122900$. No locally integrable symmetric tensor of the fixed-frame form $H(A,B,C,D)$ in (16), together with a locally integrable scalar pressure q, can satisfy in the distributional sense*

$$\operatorname{div} H + \nabla q = f, \qquad |\delta c_i(A,B,C,D)| \le \gamma\rho \quad (1 \le i \le 7).$$

Proof. Use the divergence-free meridional field v from (20). Its nonzero coupling coordinates to H are

$$a(r,z) = \partial_z v_r + \partial_r v_z = -rP\eta''' + (3P' + rP'')\eta',$$
$$d(r,z) = \partial_z v_z - \partial_r v_r = (3P + 2rP')\eta''.$$

Indeed, $H : \nabla v = Aa + Dd$; the B and C components vanish because v is meridional and independent of θ. By Proposition 5.3, integration by parts and $\rho \geq 0$ would force

$$L \leq \gamma S, \quad S = \int_1^2 \int_{-1}^1 rP(r)\eta(z) \max\left(|a|, \frac{|a|}{2} + \frac{|d|}{4}\right) \mathrm{d}z\, \mathrm{d}r. \tag{27}$$

There is an exact rational upper bound

$$S \leq \frac{5838537817787225523484624248207}{150323855360000000000000000000000} < \frac{1}{25}. \tag{28}$$

For completeness, partition $[1,2] \times [-1,1]$ into 40^2 rational rectangles. On each rectangle express the separated polynomials a and d in a common-degree bivariate Bernstein basis. The convex-hull property bounds $|a|$ and $|d|$ by the largest absolute Bernstein coefficient. Multiply $\max(a_{ij}, a_{ij}/2 + d_{ij}/4)$ by the *exact* nonnegative cell mass $\int rP\,\mathrm{d}r \int \eta\,\mathrm{d}z$, and sum the 1600 rational terms. This yields the first inequality in (28); the second is an integer comparison. The accompanying pure-rational program reproduces each coefficient and sum. Consequently

$$L - \gamma S > \frac{256}{40425} - \frac{15971}{122900 \cdot 25} = \frac{5637293}{4968232500} > 0,$$

contradicting (27). The pressure term vanishes by solenoidality of v. The zero-extended test field is compactly supported and Lipschitz, so smooth divergence-free mollifications justify the same integration-by-parts identity for locally integrable H and q. □

This obstruction concerns the *absolute correction budget* $|\delta c_i| \leq \gamma\rho$ in one fixed frame. It does not rule out every nonnegative seven-ray stress: the one-sided requirement $c_i^{(0)}\rho + \delta c_i \geq \rho/100$ admits perturbations outside that sufficient absolute budget. We now compute the exact support of this larger admissible set rather than extending the obstruction by analogy.

Proposition 5.5 (Sharp support of the one-sided seven-ray cone)**.** *Let $\rho \geq 0$ and let $\mathcal{P}_\rho$ contain precisely those (A,B,C,D) for which all seven original coefficients satisfy $c_i^{(0)}\rho + \delta c_i(A,B,C,D) \geq \rho/100$. Define*

$$g = \frac{15971}{122900}, \quad h = \frac{22371}{122900}, \quad x = A - 4B.$$

Then $\mathcal{P}_\rho$ is described exactly by

$$-\frac{g\rho}{4} \leq D \leq \frac{(g+h)\rho}{8}, \qquad |B| \leq \frac{g\rho}{8},$$
$$|x| \leq \frac{g\rho}{2} + 2D, \qquad \left|C - \frac{(h-g)\rho}{4}\right| \leq \frac{(g+h)\rho}{4} - 2D. \tag{29}$$

For arbitrary dual coefficients $(a,b,c,d) \in \mathbb{R}^4$ its support is

$$\sigma_\rho^+(a,b,c,d) = \rho\Bigg[\frac{h-g}{4}c + \frac{g}{8}|4a+b| + \max\left\{\frac{3g+h}{4}|c| - \frac{g}{4}d,\ \frac{3g+h}{4}|a| + \frac{g+h}{8}d\right\}\Bigg]. \tag{30}$$

The universal upper bound in (30) is kernel-checked in Lean under the original seven coefficient floors.

Proof. The six ordinary ray centres have available negative correction $-g\rho$ and the third ray has $-h\rho$. The paired fourth and fifth coefficient inequalities yield the bound on $|x|$; the sixth and seventh yield the bound on $|B|$. The second and third give the C interval, whose nonnegative half-width forces the upper bound on D. The half-width for x forces the lower bound on D. The remaining first-ray inequality $-3D/2 \geq -g\rho$ is then redundant, since $(g+h)/8 \leq 2g/3$. This proves the exact description, including when $\rho = 0$.

The objective is $ax + (4a+b)B + cC + dD$. For each fixed D, the independent intervals for x, B, C give a maximum equal to

$$\frac{h-g}{4}\rho c + \frac{g\rho}{8}|4a+b| + \left(\frac{g\rho}{2} + 2D\right)|a| + \left(\frac{(g+h)\rho}{4} - 2D\right)|c| + Dd.$$

This is affine in D. Substitution of its two endpoints gives (30). Signs of x, B and the displacement of C attain these extrema, establishing sharpness. □

Theorem 5.6 (Exact joint axial–azimuthal seven-ray cone)**.** *Fix $\rho \geq 0$, and retain the genuine one-sided seven-ray floor of Proposition 5.5. Given $A, C \in \mathbb{R}$, there exist $B, D \in \mathbb{R}$ satisfying all seven coefficient floors if and only if*

$$\begin{aligned} -g\rho \leq C &\leq \frac{g+h}{2}\rho, \\ |A| - C &\leq \frac{3g}{2}\rho, \\ |A| + C &\leq \left(g + \frac{h}{2}\right)\rho. \end{aligned} \tag{31}$$

The projection is a hexagon with vertices

$$\left(\pm\frac{g\rho}{2}, -g\rho\right), \qquad \left(\pm\frac{(5g+h)\rho}{4}, \frac{(h-g)\rho}{4}\right), \qquad \left(\pm\frac{g\rho}{2}, \frac{(g+h)\rho}{2}\right). \tag{32}$$

For $a, c \in \mathbb{R}$, its exact joint support is

$$\sigma_\rho^{AC}(a,c) = \rho\left[\frac{h-g}{4}c + \frac{g}{2}|a| + \frac{3g+h}{4}\max\{|a|,|c|\}\right]. \tag{33}$$

In particular, the separate sharp values of A and C cannot be attained simultaneously.

Proof. The fourth and fifth coefficient floors, together with the sixth and seventh, imply

$$|A - 4B| \leq \frac{g\rho}{2} + 2D, \qquad |4B| \leq \frac{g\rho}{2}.$$

Thus $|A| \leq g\rho + 2D$. The second and third floors give

$$2D \leq C + \frac{g\rho}{2}, \qquad 2D \leq \frac{h\rho}{2} - C.$$

The nonnegative half-width of the $A - 4B$ interval forces $D \geq -g\rho/4$, giving the two bounds on C and both mixed inequalities in (31).

Conversely suppose (31) holds and choose

$$D = \min\left\{\frac{g\rho + 2C}{4}, \frac{h\rho - 2C}{4}\right\}, \qquad 4B = \operatorname{clip}\left(A; -\frac{g\rho}{2}, \frac{g\rho}{2}\right),$$

where $\operatorname{clip}(s; l, u) = \max\{l, \min\{s, u\}\}$. The displayed inequalities give

$$-\frac{g\rho}{4} \leq D \leq \frac{(g+h)\rho}{8} \leq \frac{2g\rho}{3}, \qquad D \geq \frac{|A| - g\rho}{2},$$

and

$$|A - 4B| = \max\left\{|A| - \frac{g\rho}{2}, 0\right\} \le \frac{g\rho}{2} + 2D.$$

These relations verify all seven floors, including the first-ray inequality $-3D/2 \ge -g\rho$. They also cover $\rho = 0$. The intersection of the four sloping and two horizontal half-planes has exactly the six vertices in (32). Maximizing $aA + cC$ at those vertices yields (33). □

For the unrotated cylindrical source frame the same inequalities persist upon positive-weight angular and axial averaging. Namely, at a fixed radius let

$$S_z = \int \langle K_{rz}\rangle_\theta \, dz, \qquad S_\theta = \int \langle K_{r\theta}\rangle_\theta \, dz, \qquad M_\rho = \int \langle \rho\rangle_\theta \, dz.$$

Provided the source frame agrees with the cylindrical frame, the pair (S_z, S_θ) satisfies (31) with (A, C, ρ) replaced by (S_z, S_θ, M_ρ). The axial source reconstruction must retain the $K_{zz} + p$ boundary jump, and the azimuthal reconstruction must retain its $K_{\theta z}$ endcap. For a genuinely rotating source frame the correct necessary bound instead uses the full dual support $\sigma_\rho^+(a\lambda(E^\mathsf{T} G_z E) + c\lambda(E^\mathsf{T} G_\theta E))$ integrated over the slice, where $G_z = e_r \otimes e_z$ and $G_\theta = e_r \otimes e_\theta$; fixed-frame hexagon coefficients cannot simply be transported through a tilt.

Proposition 5.7 (The manufactured dual test is inconclusive for the one-sided cone)**.** *For the fixed diagnostic source and divergence-free test field in Proposition 5.4, let a, d be the coupling expressions there, and define*

$$J = \int_1^2 \int_{-1}^1 rP(r)\eta(z)\, \sigma_1^+(-a(r,z), 0, 0, -d(r,z)) \, \mathrm{d}z \, \mathrm{d}r.$$

An exact rational partition certificate proves

$$J \ge \frac{24989775099966161649385744300330541}{3879708382986240000000000000000000000} > \frac{256}{40425} + \frac{1}{10000}. \tag{34}$$

Consequently the necessary dual inequality $L \le J$ is not contradicted by this test field under the one-sided cone. This asserts neither the existence of a stress nor the absence of an obstruction from a different test field.

Proof. The explicit support (30), evaluated on $(-a, 0, 0, -d)$, is the maximum of the four signed polynomial actions

$$\frac{ga}{2} + \frac{gd}{4}, \quad -\frac{ga}{2} + \frac{gd}{4}, \quad \left(\frac{g}{2} + \frac{3g+h}{4}\right)a - \frac{g+h}{8}d, \quad -\left(\frac{g}{2} + \frac{3g+h}{4}\right)a - \frac{g+h}{8}d.$$

Divide the rectangle into 40×40 rational subrectangles. On each cell, the integral of the pointwise maximum is at least the maximum of the four *exact signed polynomial integrals* against the nonnegative weight $rP\eta$. Their rational sum gives the first inequality in (34). The second is an exact integer comparison. For independent error detection, the accompanying program also bounds the same integral from above with signed bivariate Bernstein coefficients. The pressure contribution vanishes because the test field is divergence free. □

The contrast is mathematically substantive: the absolute-budget obstruction is strict, whereas the exact one-sided cone is wider and this particular necessary inequality has positive slack. A primal positive-cone construction would additionally require a compact symmetric-divergence inverse with all support and derivative bounds. In addition, $f = P\eta' e_z$ is a manufactured diagnostic source, *not* the authenticated H8 residual $g - \mathbb{P}\,\mathrm{div}(\rho R_*)$, where $\mathbb{P}$ denotes the divergence-free Leray projection; the source profiles and potentially rotating physical frame required to test that residual have not been provided in the checked project. No statement here excludes a more flexible source-matched H8 inverse or decides the global unforced problem.

Proposition 5.8 (Physical-frame pullback of the one-sided dual)**.** *Let $E(x)$ be a pointwise orthogonal 3×3 source-to-physical frame, written in Cartesian components, and set $K(x) = E(x)H(A, B, C, D)(x)E(x)^{\mathsf{T}}$ with H from (16). For any physical gradient $G = \nabla v$, put $M = E^{\mathsf{T}}GE$ and define*

$$(a_E, b_E, c_E, d_E) = (M_{rz} + M_{zr},\ M_{\theta z} + M_{z\theta},\ M_{r\theta} + M_{\theta r},\ M_{zz} - M_{rr}). \tag{35}$$

Then

$$K : G = Aa_E + Bb_E + Cc_E + Dd_E. \tag{36}$$

If every original seven-ray coefficient is at least $\rho/100$ for $\rho \geq 0$, then the pointwise necessary bound is

$$-K : G \leq \sigma^+_\rho(-a_E, -b_E, -c_E, -d_E), \tag{37}$$

where σ^+ is given exactly by (30). Both (36) and (37) are kernel-checked in Lean as pointwise finite-matrix results; the algebra does not require E to be constant.

Suppose additionally that K and q solve $\operatorname{div} K + \nabla q = f$ distributionally on a periodic cell or a region where the following integration by parts is valid. For every smooth periodic or compactly supported solenoidal v,

$$\int f \cdot v \, \mathrm{d}x \leq \int \sigma^+_{\rho(x)}(-a_E(x), -b_E(x), -c_E(x), -d_E(x)) \, \mathrm{d}x. \tag{38}$$

Proof. The Frobenius identity $(EHE^{\mathsf{T}}) : G = H : (E^{\mathsf{T}}GE)$ follows by rearranging finite sums over the three matrix indices. Inserting the nine entries of H gives (36). Apply Proposition 5.5 to the negative of this dual covector to obtain (37). Finally $\int f \cdot v = -\int K : \nabla v$ because the pressure term pairs to zero with a solenoidal v, proving (38). □

The frame transfer changes the dual coordinates even when E is constant. For example, take $G = \operatorname{diag}(1, 2, 3)$ and the orthogonal meridional rotation

$$E = \begin{pmatrix} 3/5 & 0 & 4/5 \\ 0 & 1 & 0 \\ -4/5 & 0 & 3/5 \end{pmatrix}.$$

Formula (35) gives $(a_E, b_E, c_E, d_E) = (-48/25, 0, 0, -14/25)$, whereas the frozen frame gives $(0, 0, 0, 2)$. Thus a fixed-frame numerical obstruction cannot be transferred to a varying frame without recomputing its dual integral.

For a differentiable frame, the analytic divergence additionally contains the connection terms

$$\partial_j K_{ij} = E_{ia}E_{jb}\partial_j H_{ab} + (\partial_j E_{ia})H_{ab}E_{jb} + E_{ia}H_{ab}(\partial_j E_{jb}), \tag{39}$$

with repeated indices summed. Orthogonality ensures that a source-frame scalar multiple of the identity remains a physical pressure gauge, even if E varies. Equation (39) shows why the pointwise dual theorem alone does not control the weighted derivatives of a source-matched inverse. The authenticated functions g, ρ, R_* and E, together with their support and derivative budgets, remain required to evaluate (38) on the genuine H8 residual.

Proposition 5.9 (Exact pressure-compensated annular H8 gate)**.** *Fix a time and a cylindrical annulus $a < r < b$, $c < z < d$. Let K be a smooth symmetric physical tensor, p a smooth scalar, and $f = \operatorname{div} K + \nabla p$. Denote the azimuthal average by $\langle \cdot \rangle_\theta$ and set*

$$S(r) = \int_c^d \langle K_{rz} \rangle_\theta \, \mathrm{d}z, \qquad F(r) = \int_c^d \langle f_z \rangle_\theta \, \mathrm{d}z,$$
$$q_{\mathrm{s}}(r) = \left[\langle K_{zz} + p \rangle_\theta \right]_{z=c}^{z=d}, \qquad Q(r) = \frac{1}{r} \int_a^r s \, q_{\mathrm{s}}(s) \, \mathrm{d}s. \tag{40}$$

Then the exact cylindrical identity is

$$J_f(r) := \frac{1}{r}\int_a^r sF(s)\,\mathrm{d}s = S(r) - \frac{a}{r}S(a) + Q(r). \tag{41}$$

Suppose $K = EH(A,B,C,D)E^{\mathsf{T}}$, where $E(r,\theta,z)$ is orthogonal and the seven H8 ray coefficients exceed $\rho/100$ for $\rho \geq 0$. Let G be the physical matrix having $G_{rz} = 1$ and all other entries zero, and write λ_E for the four coordinates of (35) applied to this G. Define the exact slice supports

$$\begin{aligned}\mathcal{B}_{\pm}(r) &= \int_c^d \left\langle \sigma^{+}_{\rho(r,\theta,z)}(\pm\lambda_E(r,\theta,z)) \right\rangle_\theta \mathrm{d}z,\\ \mathcal{B}_{\mathrm{abs}}(r) &= \int_c^d \left\langle \sigma_{g\rho(r,\theta,z)}(\lambda_E(r,\theta,z)) \right\rangle_\theta \mathrm{d}z, \qquad g = \frac{15971}{122900}.\end{aligned} \tag{42}$$

The original absolute coefficient budget implies

$$\left| J_f(r) + \frac{a}{r}S(a) - Q(r) \right| \leq \mathcal{B}_{\mathrm{abs}}(r), \tag{43}$$

while the larger one-sided positive-coefficient cone requires only

$$-\mathcal{B}_-(r) \leq J_f(r) + \frac{a}{r}S(a) - Q(r) \leq \mathcal{B}_+(r). \tag{44}$$

These bounds hold for a spatially varying orthogonal frame; they are necessary conditions and do not construct a differential inverse.

Proof. The azimuthal mean removes the $r^{-1}\partial_\theta K_{\theta z}$ term in the cylindrical divergence. The axial component therefore obeys $F = S' + S/r + q_{\mathrm{s}}$, proving $(rS)' = r(F - q_{\mathrm{s}})$ and (41) by integration. At each spatial point $K_{rz} = K : G = \lambda_E \cdot (A,B,C,D)$. The sharp one-sided support formula (30) gives the upper and lower bounds on this quantity, and (26) gives the symmetric absolute bound. Azimuthal and axial integration followed by (41) proves both inequalities. □

Proposition 5.10 (Case 006 source-native axial interval)**.** *Work in the physical cylindrical source frame of the Case 006 seven-ray centre, for which $(R_*)_{rz} = \alpha = -4443/9832$. Let T be a smooth prepared symmetric stress, let $\rho \geq 0$, and write*

$$f = \operatorname{div}(\mathsf{T} - \rho R_*) + \nabla p_0, \qquad \mathcal{T}(r) = \int_c^d \langle \mathsf{T}_{rz}\rangle_\theta \,\mathrm{d}z, \qquad \mathcal{R}(r) = \int_c^d \langle \rho\rangle_\theta \,\mathrm{d}z.$$

Set $q_0(r) = [\langle (\mathsf{T} - \rho R_*)_{zz} + p_0\rangle_\theta]_{z=c}^{z=d}$, and let J_f be the source primitive in (41). Then

$$J_f(r) = \mathcal{T}(r) - \alpha\mathcal{R}(r) - \frac{a}{r}(\mathcal{T}(a) - \alpha\mathcal{R}(a)) + \frac{1}{r}\int_a^r sq_0(s)\,\mathrm{d}s. \tag{45}$$

Suppose that the source and the sought correction admit the corresponding compact representatives, with zero inner shear and axial pressure/tensor boundary traces. If the correction $H(A,B,C,D)$ obeys the original absolute *seven-coefficient budget $|\delta c_i| \leq \kappa\rho$, $\kappa = 15971/122900$, then*

$$|\mathcal{T}(r) - \alpha\mathcal{R}(r)| \leq \kappa\mathcal{R}(r).$$

In particular, whenever $\mathcal{R}(r) > 0$,

$$-\frac{143017}{245800} \leq \frac{\mathcal{T}(r)}{\mathcal{R}(r)} \leq -\frac{79133}{245800}. \tag{46}$$

If the angular coefficient $B = H_{\theta z}$ has already been selected and $\mathcal{B}(r) = \int_c^d \langle B\rangle_\theta \, \mathrm{d}z$, the stronger condition is $|\mathcal{T} - \alpha\mathcal{R} - 4\mathcal{B}| \le (\kappa/2)\mathcal{R}$.

This interval is not *implied by the larger one-sided positive-coefficient cone. With $g = 15971/122900$ and $h = 22371/122900$, its sharp physical rz maximum in this same frame is*

$$A_{\max} = \frac{5g+h}{4}\rho = \frac{51113}{245800}\rho > \kappa\rho.$$

Indeed $D = (g+h)\rho/8$, $B = g\rho/8$, $C = (h-g)\rho/4$ and $A = (5g+h)\rho/4$ preserve all seven coefficient floors at or above $\rho/100$. A spatially rotated source frame additionally requires the transported support in (42); neither scalar interval may simply be copied to an arbitrary rotating frame.

Proof. Equation (45) follows by applying Proposition 5.9 to $K = \mathsf{T} - \rho R_*$ and using $(R_*)_{rz} = \alpha$. Under the stated compact boundary assumptions the source primitive is the axial integral of the correction shear $H_{rz} = A$. The original coefficient norm is

$$\max\{8|B|,\ 2|C| + 4|D|,\ 2|A - 4B| + 4|D|\} \le \kappa\rho.$$

It follows that $|A| \le 4|B| + |A - 4B| \le \kappa\rho$ and $|A - 4B| \le \kappa\rho/2$. Integrating over the slice proves the two necessary bounds. Exact subtraction of the rational centre and margin gives $\alpha - \kappa = -143017/245800$ and $\alpha + \kappa = -79133/245800$. Finally the displayed one-sided witness satisfies the seven original coefficient floors by direct substitution in their rational formulas; its A strictly exceeds $\kappa\rho$. □

The boundary primitive cannot be discarded in a local test. For example, on $1 \le r \le 2$, $0 \le z \le 1$, take $K = 0$ and $p = z$. Then $F = q_{\mathrm{s}} = 1$ and $S = 0$, whereas $J_f(r) = (r^2 - 1)/(2r)$ is nonzero. The zero stress satisfies either cone with $\rho = 0$. Thus $|J_f| \le \mathcal{B}$ without the boundary correction would give a false obstruction. This example is local and does not assert compact pressure support.

For a finite exact check, let the frame be independent of z, θ on each radial slice and write $\bar\rho(r) = \int_c^d \rho(r,z)\,\mathrm{d}z$. Then the supports in (42) reduce to their respective finite-dimensional formulas at weight $\bar\rho$. For the rationally rotating frame

$$E(r) = \begin{pmatrix} \frac{1-t^2}{1+t^2} & 0 & \frac{2t}{1+t^2} \\ 0 & 1 & 0 \\ -\frac{2t}{1+t^2} & 0 & \frac{1-t^2}{1+t^2} \end{pmatrix}, \qquad t = r - 1,$$

the physical rz shear has pulled-back covector $\lambda_E = (-7/25, 0, 0, 24/25)$ at $r = 3/2$. Writing $h = 22371/122900$, exact evaluation gives

$$\mathcal{B}_+ = \frac{19h + 47g}{100}\bar\rho, \qquad \mathcal{B}_- = \frac{19g}{50}\bar\rho, \qquad \mathcal{B}_{\mathrm{abs}} = \frac{19g}{50}\bar\rho. \tag{47}$$

The asymmetry is real: replacing the original absolute budget by one-sided positivity can change a failed test into an inconclusive one. These rational point evaluations are not uniform estimates for a smooth shell, and no source authenticity follows from them.

Proposition 5.11 (Source-shaped meridional flux, including viscosity)**.** *Let $U = (U_r, U_\theta, U_z)$ be an axisymmetric divergence-free smooth velocity on the cylindrical slice. Consider finitely many poloidal stream functions*

$$\Psi_i(r,z) = K_i a_i(r) b(r) \Xi_i(z), \qquad P_{r,i} = -r^{-1}\partial_z\Psi_i, \quad P_{z,i} = r^{-1}\partial_r\Psi_i,$$

where K_i are constant and $\Xi_i = \Xi_i' = 0$ at both axial endpoints. Put $P = \sum_i P_i$ and allow any purely azimuthal toroidal field $T_\theta e_\theta$. Define the axial moments

$$I_i(r) = \int_c^d U_r(r,z)\Xi_i(z)\,\mathrm{d}z, \quad m_i = \int_c^d \Xi_i(z)\,\mathrm{d}z, \quad J_{ij} = \int_c^d \Xi_i'(z)\Xi_j(z)\,\mathrm{d}z.$$

Then the axial integral of the physical quadratic rz flux increment is

$$\begin{aligned}
\mathcal{C}(r) &= \int_c^d \left[(U+P+T)_r(U+P+T)_z - U_rU_z\right]\mathrm{d}z = \Gamma(r)b'(r) + \Delta(r)b(r) + Q_2(r)b(r)^2,\\
\Gamma &= \frac{1}{r}\sum_i K_i a_i I_i, \qquad \Delta = \frac{1}{r}\sum_i K_i\left[a_i' I_i - a_i\left(I_i' + \frac{I_i}{r}\right)\right],\\
Q_2 &= \frac{1}{r^2}\sum_{i<j} K_iK_j(a_ja_i' - a_ia_j')J_{ij}.
\end{aligned} \tag{48}$$

The contribution to the same flux from the canonical Newtonian viscous stress of P is exactly

$$\mathcal{V}_\nu(r) = -\nu\int_c^d (\partial_z P_r + \partial_r P_z)\,\mathrm{d}z = -\nu\sum_i K_i m_i \frac{d}{dr}\left(\frac{(a_ib)'}{r}\right). \tag{49}$$

In particular its coefficient of b'' is $-\nu r^{-1}\sum_i K_i m_i a_i$; it cannot be included in a $O(b^2)$ remainder merely by assigning a flat radial cutoff.

Proof. The mixed core–shaper product is $r^{-1}\sum_i K_i[(a_ib)'I_i - a_ib\int_c^d U_z\Xi_i'\,\mathrm{d}z]$. Axial integration by parts and $r^{-1}\partial_r(rU_r) + \partial_zU_z = 0$ give $\int U_z\Xi_i' = I_i' + I_i/r$, proving the linear terms. The self-products vanish because $\int \Xi_i'\Xi_i = 0$. For $i \neq j$ integration by parts gives $J_{ji} = -J_{ij}$; the paired cross-products reduce to $r^{-2}K_iK_j(a_ja_i' - a_ia_j')b^2J_{ij}$. All rz components of a purely toroidal velocity vanish. Finally the integral of ∂_zP_r vanishes by the endpoint conditions and $\int \partial_rP_z = \sum_i K_im_i\partial_r((a_ib)'/r)$. This proves the viscous identity and its displayed b'' coefficient. □

For an exact local diagnostic take $U_r = -2r$, $U_z = 4z$, $c = -1$, $d = 1$, $b = (r-1)^3$, $\nu = 1$, and two shapers

$$(K_1, a_1, \Xi_1) = (1, 1, (1-z^2)^2), \qquad (K_2, a_2, \Xi_2) = (2, r, z(1-z^2)^2).$$

At $r = 2$, direct polynomial integration and the reduced formula independently give

$$\Gamma = -\frac{32}{15}, \quad \Delta = \frac{32}{15}, \quad Q_2 = \frac{64}{315}, \quad \mathcal{V}_1 = -\frac{12}{5}, \quad \mathcal{C} + \mathcal{V}_1 = -\frac{2036}{315}. \tag{50}$$

The polynomial functions illustrate the exact algebra on a local slice; their zero extensions are not C^∞ compact H8 cutoffs. Neither the still-free localization stresses nor the common pressure have been inserted into this number.

Proposition 5.12 (Flat-edge second-derivative obstruction)**.** *Under the hypotheses of Proposition 5.11, put $M(r) = \sum_i K_i m_i a_i(r)$. The shaper-induced kinetic and viscous stress-shear increment has the exact decomposition*

$$\begin{aligned} \mathcal{C} + \mathcal{V}_\nu &= \kappa_2 b'' + \kappa_1 b' + \kappa_0 b + \kappa_q b^2, \\ \kappa_2 &= -\frac{\nu M}{r}, & \kappa_1 &= \Gamma - \nu\left(\frac{2M'}{r} - \frac{M}{r^2}\right), \\ \kappa_0 &= \Delta - \nu\left(\frac{M''}{r} - \frac{M'}{r^2}\right), & \kappa_q &= Q_2. \end{aligned} \tag{51}$$

Let $r_0 > 0$ and $I = [r_0, r_0 + \delta]$, and suppose these four coefficients are bounded on I with $\inf_I |\kappa_2| > 0$. If $b \in C^2(I)$ satisfies $b(r_0) = b'(r_0) = 0$ and another stress-shear contribution R obeys, for finite constants $L, L' \geq 0$,

$$|R| \leq L(|b| + |b'| + b^2), \qquad |\mathcal{C} + \mathcal{V}_\nu + R| \leq L' b^2 \quad \text{on } I, \tag{52}$$

then $b \equiv 0$ throughout I.

Proof. Differentiating $M'b + Mb'$ divided by r in (49) yields $Mb''/r + (2M'/r - M/r^2)b' + (M''/r - M'/r^2)b$, which proves (51). Let $\eta = \inf_I |\kappa_2| > 0$ and $L_b = \sup_I |b| < \infty$. The two bounds in (52), boundedness of the other three coefficients and $b^2 \leq L_b|b|$ yield a constant $C < \infty$ such that

$$|b''(r)| \leq C(|b(r)| + |b'(r)|) \quad (r \in I).$$

By the initial conditions and the fundamental theorem of calculus, the continuous nonnegative function $w = |b| + |b'|$ satisfies $w(r) \leq (1 + C)\int_{r_0}^r w(s)\,\mathrm{d}s$. Gronwall's elementary integral inequality gives $w \equiv 0$, and therefore $b \equiv 0$. □

For the two profiles in (50), direct integration gives $m_1 = 16/15$ and $m_2 = 0$; hence $M = 16/15$ and

$$\mathcal{C} + \mathcal{V}_1 = -\frac{16}{15r}b'' + \left(-\frac{32}{15} + \frac{16}{15r^2}\right)b' + \frac{64}{15r}b + \frac{256}{315r^2}b^2. \tag{53}$$

At $r = 2$ and $b = (r-1)^3$ this again gives $-2036/315$. Since $\kappa_2 = -16/(15r)$ is nowhere zero on any annulus with $r > 0$ (indeed $|\kappa_2| \geq 8/15$ on $1 \leq r \leq 2$), Proposition 5.12 prohibits a nontrivial one-sided flat cutoff from satisfying an $O(b^2)$ budget for this *restricted* flux whenever its remaining terms obey the first inequality of (52).

In the H8 application, the two slice support functions in (42) yield such a target bound only if the relevant reference-centre-subtracted shear is precisely the quantity in (52) and both supports are $O(b^2)$ on the edge collar. Neither follows from the source architecture alone. Nontrivial compactness can instead require a vanishing effective b'' coefficient, a compensating transition or overlap term of second-jet size, a pressure boundary primitive with the requisite source matching, or a density budget larger than $O(b^2)$. The lemma does not rule out these possibilities and is not a contradiction for the authentic H8 residual.

The Case 006 source specifies a compact curl localization $u_b = \nabla \times (\chi A_c)$, separable poloidal and toroidal shell shapers, and a toroidal pressure radial derivative, but does not fix the smooth profiles, the complete four-patch interactions, or a common pressure representative. Hence no value of (41) has yet been certified for the complete H8 residual. The exact code records the transition, shaper, viscosity, overlap, reference-centre and axial boundary channels separately; omitting any named channel blocks evaluation. Supplying zero for a channel requires mathematical source justification, not merely successful execution of the code.

Proposition 5.13 (Quarter-turn localization and its nonlinear overlaps)**.** *Let $Q(x,y,z) = (-y,x,z)$ and χ be invariant under Q. On a simply connected Cartesian chart let A be a smooth vector potential, and denote the active push-forward by $Q_*v(x) = Qv(Q^{-1}x)$. Write*

$$v_j = Q_*^j(\nabla \times (\chi A)), \quad 0 \le j \le 3, \qquad u_b = \frac{1}{4}\sum_{j=0}^{3} v_j.$$

Then u_b is solenoidal and equals the curl of the normalized quarter-turn average of χA. Its full kinetic stress splits exactly into

$$u_b \otimes u_b = \frac{1}{16}\sum_{j=0}^{3} v_j \otimes v_j + \frac{1}{16}\sum_{\substack{0\le j,k\le 3\\ j\neq k}} v_j \otimes v_k. \tag{54}$$

For the material derivative, the same diagonal and ordered-cross decomposition holds with $v_j \otimes v_k$ replaced by $(v_j \cdot \nabla)v_k$. The twelve cross products do not generally vanish under fourfold symmetry.

Proof. Rigid rotations commute with curl, and the curl of an average is the average of the curls. Each v_j is divergence free. Expanding the product of the two sums for u_b proves (54); applying the Leibniz bilinearity of directional differentiation proves the derivative formula. The normalization $1/16$ follows from two factors of $1/4$ and cannot be dropped. □

For example, on a local chart take $\chi = 1$ and $A = (-yz, xz, 0)$. Every rotated field is $v_j = (-x,-y,2z)$, so in cylindrical notation $u_r = -r$, $u_z = 2z$, and $u_r u_z = -2rz$. The diagonal terms in (54) contribute $-rz/2$ to this shear, whereas the ordered overlaps contribute $-3rz/2$. Thus three quarters of the core shear is carried by the overlap partition even for an exactly fourfold-equivariant field. This local polynomial example does not claim a compact source profile; spatial disjointness, if asserted for different carrier supports, must be checked separately.

Proposition 5.14 (Shared-envelope second jet and core circulation)**.** *Let $r_0 < r < r_1$ be a cylindrical annulus separated from the axis, with angular coordinate θ and fixed axial endpoints $z_- < z_+$. Let $\mathcal{A} = (\mathcal{A}_r, \mathcal{A}_\theta, \mathcal{A}_z)$ be a smooth, 2π-periodic vector potential in the cylindrical orthonormal frame (possibly only C_4-equivariant), and let $U_c = \nabla \times \mathcal{A}$. Suppose the localized core and the poloidal stream functions of Proposition 5.11 share one radial envelope $b(r)$, so their total meridional correction is obtained from*

$$\nabla \times \left[b(r)\left(\mathcal{A} + \sum_i \frac{K_i a_i(r)\Xi_i(z)}{r} e_\theta \right)\right].$$

An arbitrary smooth background independent of b may be included. Denote the full angular mean by $\langle\cdot\rangle_\theta$, set

$$M(r) = \sum_i K_i m_i a_i(r), \qquad V(r) = \int_{z_-}^{z_+} \langle (U_c)_z(r,\theta,z)\rangle_\theta \, \mathrm{d}z,$$

and let $S_b(r)$ be the axial/angular integral of the increment in the physical shear $K_{rz} = u_r u_z - \nu(\partial_z u_r + \partial_r u_z)$. The coefficient of $b''(r)$ in S_b is exactly

$$\kappa_{2,\mathrm{eff}}(r) = -\nu\left[\int_{z_-}^{z_+} \langle \mathcal{A}_\theta(r,\theta,z)\rangle_\theta \, \mathrm{d}z + \frac{M(r)}{r}\right]. \tag{55}$$

All the remaining terms in S_b are expressions in b, b' with smooth bounded coefficients on compact subannuli. In particular, the quadratic overlaps among the four velocities contain no b'' term in this physical shear *channel.*

For any reference radius $r_* \in [r_0, r_1)$, put

$$H_* := r_* \int_{z_-}^{z_+} \langle \mathcal{A}_\theta(r_*, \theta, z) \rangle_\theta \, \mathrm{d}z.$$

Then (55) has the potential-free expression

$$\kappa_{2,\mathrm{eff}}(r) = -\frac{\nu}{r} \left[H_* + \int_{r_*}^{r} sV(s) \, \mathrm{d}s + M(r) \right]. \tag{56}$$

Consequently, identical *second-jet cancellation on an interval requires and is implied by a single anchor equality $H_* + \int_{r_*}^{r_a} sV(s) \, \mathrm{d}s + M(r_a) = 0$ and the differential identity*

$$rV(r) + M'(r) = 0 \tag{57}$$

throughout that interval. If the potential extends smoothly across the axis on complete disks, one may set $r_* = 0$ and $H_* = 0$.

Proof. Writing $A_\theta^{\mathrm{tot}} = \mathcal{A}_\theta + \sum_i K_i a_i \Xi_i / r$, the cylindrical curl gives

$$u_r - U_r = b \left(\frac{1}{r} \partial_\theta \mathcal{A}_z - \partial_z A_\theta^{\mathrm{tot}} \right), \qquad u_z - U_z = b \left(\frac{1}{r} \partial_r (r A_\theta^{\mathrm{tot}}) - \frac{1}{r} \partial_\theta \mathcal{A}_r \right) + b' A_\theta^{\mathrm{tot}}.$$

The quadratic product $u_r u_z$ therefore uses at most b and b'. In the viscous shear only $\partial_r(u_z - U_z)$ produces b'', with coefficient $-\nu A_\theta^{\mathrm{tot}}$. Angular/axial integration proves (55). Meanwhile

$$(U_c)_z = \frac{1}{r} \partial_r (r \mathcal{A}_\theta) - \frac{1}{r} \partial_\theta \mathcal{A}_r.$$

The full angular average annihilates the last derivative; hence $\partial_r (r \int \langle \mathcal{A}_\theta \rangle_\theta \, \mathrm{d}z) = rV(r)$. Integrating from r_* proves (56). Differentiating its bracket proves the stated characterization of identical cancellation. Smooth axis extension forces $r \langle \mathcal{A}_\theta \rangle_\theta \to 0$ as $r \downarrow 0$. □

The annular reference circulation matters. For the single profile $(K, a, \Xi) = (1, 1, (1 - z^2)^2)$ on $[-1, 1]$, one has $M = 16/15$. Choosing the local, C_4-invariant azimuthal potential $\mathcal{A}_\theta = -8/(15r)$ gives $\int_{-1}^{1} \mathcal{A}_\theta \, \mathrm{d}z = -16/(15r)$ and $\kappa_{2,\mathrm{eff}} \equiv 0$. Here $(U_c)_z = 0$ on the annulus, while $H_* = -16/15$ supplies the cancellation. This is an algebraic counterexample to deleting the circulation constant, not an admissible axis-spanning H8 core. If a smooth axis-spanning potential is prescribed, its core axial flux rather than a free annular circulation must supply the needed moment.

The proposition concerns the canonical physical rz stress. The pressure-compensated source relation (41) retains its independent axial jump; neither pressure nor an H8 inverse stress is reconstructed here. If the actual transition cutoff χ differs from the shaper envelope b, the core contributions cannot be assigned to the same b'' coefficient without an additional source identity. Thus the genuine H8 test still requires the core flux V, the reference circulation, the shared-cutoff decision, and the full source-matched cone density and frame.

Proposition 5.15 (Independent core and shaper cutoffs)**.** *On an annular cylindrical patch let $\mathcal{A}$ be any smooth, 2π-periodic vector potential, let $\chi = \chi(r, \theta, z)$ be a smooth core cutoff, and let $b = b(r)$ be an* independent *radial poloidal cutoff. Put*

$$P_\theta(r, z) := \sum_i \frac{K_i a_i(r) \Xi_i(z)}{r}, \qquad u = U + \nabla \times (\chi \mathcal{A} + b P_\theta e_\theta),$$

*where U is a fixed smooth background. If $S(r)$ is the axial/angular integral of the physical shear $K_{rz} = u_r u_z - \nu(\partial_z u_r + \partial_r u_z)$, its full cutoff-*second*-derivative part is*

$$S_{\mathrm{jet},2}(r) = -\nu \int_{z_-}^{z_+} \left\langle \mathcal{A}_\theta(\chi_{rr} - \chi_{zz}) + \frac{\mathcal{A}_z}{r}\chi_{\theta z} - \frac{\mathcal{A}_r}{r}\chi_{r\theta} + P_\theta b'' \right\rangle_\theta \mathrm{d}z. \tag{58}$$

The difference $S - S_{\mathrm{jet},2}$ contains the cutoffs and their first derivatives, but no second derivative of either cutoff. The nonlinear cross terms, including all ordered C_4 core overlaps, belong to this remainder. In particular, a radial shaper b and a genuinely three-variable χ do not have one common second-derivative coefficient.

For any specified pressure representative p, the complete annular spatial-divergence *source still obeys*

$$q_{\mathrm{s}}(r) = [\langle u_z^2 - 2\nu\partial_z u_z + p\rangle_\theta]_{z_-}^{z_+}, \qquad J_f = S - \frac{r_a}{r}S(r_a) + \frac{1}{r}\int_{r_a}^{r} s q_{\mathrm{s}}(s)\,\mathrm{d}s.$$

For the full unforced Navier–Stokes residual $\mathcal{N}(u,p) = \partial_t u + \operatorname{div} K + \nabla p$, one must also keep

$$T_z(r) = \int_{z_-}^{z_+} \langle \partial_t u_z\rangle_\theta \,\mathrm{d}z, \qquad J_{\mathcal{N}}(r) = J_f(r) + \frac{1}{r}\int_{r_a}^{r} s T_z(s)\,\mathrm{d}s. \tag{59}$$

No smallness of the pressure primitive or the first-derivative remainder follows from (58); neither T_z nor its primitive may be set to zero without a source identity.

Proof. The cylindrical curl identities give

$$u_r - U_r = \frac{1}{r}\partial_\theta(\chi\mathcal{A}_z) - \partial_z(\chi\mathcal{A}_\theta + bP_\theta),$$
$$u_z - U_z = \frac{1}{r}\partial_r\big(r(\chi\mathcal{A}_\theta + bP_\theta)\big) - \frac{1}{r}\partial_\theta(\chi\mathcal{A}_r).$$

Each velocity contains at most one derivative of either cutoff; therefore $u_r u_z$ contains no cutoff second derivative. Differentiating the first line in z and the second in r produces exactly the four terms in (58). The rest contains only lower cutoff jets. The pressure identity is the integrated axial component of $\operatorname{div} K + \nabla p = f$ and does not identify an independent radial primitive of p with a time-only gauge. Adding the axial marginal of $\partial_t u$ and integrating the radial first-order equation proves (59). □

The source distinction is substantive: the available Case 006 structured-stress manuscript prescribes the core cutoff as a C_4-invariant function of similarity coordinates, equal to one on an inner subcore, whereas its poloidal shaper uses a separately specified shell profile $\Phi(r/\ell)\Xi(z/c)$. Those definitions neither identify the two radial envelopes nor provide a complete frozen four-patch source. Equation (58) is therefore the appropriate general decomposition; the shared-envelope Proposition 5.14 is its additional-hypothesis specialization.

Proposition 5.16 (Variable-frame dual enclosure with angular test)**.** *Consider the one-sided seven-ray coefficient cone and its trace-free representative H in (16). For every nonnegative density ρ, the pointwise coefficient floors imply*

$$|A|, |B|, |C|, |D| \le \frac{\rho}{4}, \qquad \|H\|_F \le \rho.$$

Let a cylindrical shear-test slice be partitioned into finitely many measurable cells Ω_j. On each cell suppose the physical orthogonal frame $E(x)$ admits a fixed rational orthogonal frame E_j and a certified *bound*

$$\sup_{\Omega_j} \|E(x) - E_j\|_F \le \delta_j, \qquad m_j = \int_{\Omega_j} \rho(x)\,\mathrm{d}\mu(x).$$

Here $\mathrm{d}\mu = \mathrm{d}z\ \mathrm{d}\theta/(2\pi)$ *on the slice. If* $\sigma^+(\lambda)$ *is the exact unit-density one-sided support and* $G(x) = e_r(\theta) \otimes e_z$*, define*

$$\widehat{B}_\pm = \sum_j \int_{\Omega_j} \rho(x)\, \sigma^+\Big(\pm\lambda(E_j^{\mathsf{T}} G(x) E_j)\Big)\ \mathrm{d}\mu(x), \qquad \varepsilon_E = 2\sum_j m_j \delta_j.$$

Then the genuine spatially varying shear supports satisfy

$$\max(0, \widehat{B}_\pm - \varepsilon_E) \le B_\pm \le \widehat{B}_\pm + \varepsilon_E. \tag{60}$$

These bounds remain valid for smooth nonanalytic frames and densities, provided the weighted cell support integrals and frame-error bounds are rigorously enclosed. Alternatively the integrals may be bounded by m_j *times certified cellwise lower and upper support values. Sampling a frame and a physical test at cell centres alone does not provide such a certificate.*

Proof. Applying the exact support formula of Proposition 5.5 to the eight signed coordinate covectors gives the largest respective normalized absolute extrema

$$\frac{51113}{245800}, \quad \frac{15971}{983200}, \quad \frac{19171}{122900}, \quad \frac{19171}{491600}.$$

Each is strictly less than $1/4$, so $\|H\|_F^2 = 2(A^2 + B^2 + C^2 + D^2) \le \rho^2/2$. For orthogonal E, E_j,

$$EHE^{\mathsf{T}} - E_j H E_j^{\mathsf{T}} = (E - E_j)HE^{\mathsf{T}} + E_j H(E - E_j)^{\mathsf{T}}.$$

The Frobenius pairing with the unit-norm $G(x)$ consequently changes by at most $2\rho\|E - E_j\|_F$. Taking the pointwise supremum over the coefficient cone, retaining $G(x)$ in the frozen-frame support, and then integrating over each cell proves (60). □

An earlier cell-mass-only expression which omitted the integral in $\widehat{B}_\pm$ is invalid for physical Cartesian reference frames. Indeed, even if $E(x) = E_j = I$ and $\delta_j = 0$, its test tensor $G(\theta)$ varies. At $\theta = 0$ the unit-density support is $51113/245800$, while at $\theta = \pi/2$ it is $15971/983200$. Thus zero frame error does not imply zero angular-test error. A piecewise-constant-test implementation is valid only with an additional certified bound $\eta_j \ge \sup_{\Omega_j}\|G(x) - G_j\|_F$, in which case $\sum_j m_j\eta_j$ is added to ε_E.

The dual bound controls values of the rotating stress, not its divergence. In a Cartesian chart, $K = EHE^{\mathsf{T}}$ obeys

$$\partial_j K_{ij} = E_{ia}E_{jb}\partial_j H_{ab} + ((\partial_j E_{ia})E_{jb} + E_{ia}(\partial_j E_{jb}))H_{ab}, \tag{61}$$

with repeated indices summed. Thus even constant H can generate a nonzero source when E rotates. Cylindrical components require their geometric connection terms as well.

The angular form of this connection is now an exact Lean theorem. More generally, let $E(t)$ be a differentiable orthogonal 3×3 frame, $H(t)$ a differentiable coefficient matrix, and $\Omega(t) = E(t)^{\mathsf{T}}E'(t)$. Differentiation of $E^{\mathsf{T}}E = I$ gives $\Omega^{\mathsf{T}} = -\Omega$. The ordinary matrix product rule and $E' = E\Omega$ yield

$$\frac{d}{dt}(EHE^{\mathsf{T}}) = E(H' + \Omega H - H\Omega)E^{\mathsf{T}}. \tag{62}$$

The `FrameConnectionDerivative` module proves each scalar physical-tensor entry derivative from explicit entrywise derivative hypotheses, proves the skew and commutator identities under row orthogonality and differentiated orthogonality, and checks an actual trigonometric planar rotation. For $H = \mathrm{diag}(1, 2, 0)$, the $(0, 1)$ entry changes at rate -1 at zero angle. This verifies the sign of (62) and refutes omission of its commutator even for a constant frame-coordinate stress.

Proposition 5.17 (Complete moving-frame source and axial trace)**.** *Let $E(x) \in SO(3)$ be a differentiable physical frame on a Cartesian chart, and let $H = H^{\mathsf{T}}$ be its differentiable stress-coordinate matrix. Write $\Gamma_i = E^{\mathsf{T}}\partial_i E$, where the index i denotes a Cartesian derivative, and set $f = \nabla \cdot (EHE^{\mathsf{T}}) + \nabla p$. Then, for each moving-frame component b,*

$$(E^{\mathsf{T}} f)_b = \sum_{i,a} E_{ia}\big(\partial_i H + \Gamma_i H - H\Gamma_i\big)_{ab} + \sum_i E_{ib}\partial_i p. \tag{63}$$

For the standard cylindrical *frame $E = (e_r, e_\theta, e_z)$ and the four-field stress in (16), the corresponding components are*

$$f_r = -\partial_r D + r^{-1}\partial_\theta C + \partial_z A - r^{-1}D + \partial_r p, \tag{64}$$
$$f_\theta = \partial_r C + 2r^{-1}C + \partial_z B + r^{-1}\partial_\theta p, \tag{65}$$
$$f_z = \partial_r A + r^{-1}A + r^{-1}\partial_\theta B + \partial_z(D + p). \tag{66}$$

If A, B, D, p are 2π-periodic in θ, have the indicated derivatives and axial traces at c, d, then

$$\int_0^{2\pi}\int_c^d r f_z \,\mathrm{d}z\,\mathrm{d}\theta = \partial_r\left(r\int_0^{2\pi}\int_c^d A \,\mathrm{d}z\,\mathrm{d}\theta\right) + r\int_0^{2\pi}[D + p]_c^d \,\mathrm{d}\theta. \tag{67}$$

In particular the pressure jump generally survives axial integration. None of these identities identifies the possibly tilted spatial frame of the unfrozen H8 construction with the standard cylindrical frame.

Proof. Orthogonality implies $\Gamma_i^{\mathsf{T}} = -\Gamma_i$ and hence $\partial_i(EHE^{\mathsf{T}}) = E(\partial_i H + \Gamma_i H - H\Gamma_i)E^{\mathsf{T}}$. Contract the Cartesian derivative index against the first tensor index and transform the remaining vector index by E^{T} to obtain (63). For the cylindrical frame,

$$E^{\mathsf{T}}\partial_\theta E = \begin{pmatrix} 0 & -1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}, \qquad \partial_r E = \partial_z E = 0.$$

Substitution gives (64)–(66). Integrating the last identity, periodicity cancels the angular derivative and the fundamental theorem of calculus produces the *combined* stress-pressure axial trace. □

The exact one-sided seven-ray floor also gives the sharp local necessary bound

$$|H_{rz}| = |A| \le \frac{51113}{245800}\,\rho. \tag{68}$$

This constant is attained by an exact rational ray allocation, and agrees with Proposition 5.16. Consequently, any source-matched construction in the frozen physical frame must simultaneously obey (67) and (68). For a spatially tilted frame the latter inequality remains in the *local* four-field coordinates, while (63) replaces the cylindrical expression. The symbolic verification accompanying this manuscript tests all three Cartesian divergence components, a nonconstant spatial tilt, and a nonzero axial pressure jump. No authentic H8 profile tuple has been certified by these tests, and the compact source-matched positive-stress inverse remains an open analytic problem.

Corollary 5.18 (Sharp azimuthal torque-flux compatibility)**.** *Under the cylindrical-frame hypotheses of Proposition 5.17, define*

$$\mathfrak{c}(r) = \int_c^d \langle C\rangle_\theta \,\mathrm{d}z, \qquad \mathfrak{b}(r) = [\langle B\rangle_\theta]_c^d, \qquad \mathfrak{f}_\theta(r) = \int_c^d \langle f_\theta\rangle_\theta \,\mathrm{d}z,$$

where $\langle\cdot\rangle_\theta$ is the normalized full-circle angular average. Then

$$\mathfrak{f}_\theta(r) = \frac{1}{r^2}\partial_r(r^2\mathfrak{c}(r)) + \mathfrak{b}(r), \tag{69}$$

and for $a > 0$,

$$\mathfrak{c}(r) = \frac{a^2}{r^2}\mathfrak{c}(a) + \frac{1}{r^2}\int_a^r s^2(\mathfrak{f}_\theta(s) - \mathfrak{b}(s))\,\mathrm{d}s. \tag{70}$$

If the original one-sided seven-ray positivity floor holds at every point and $\rho \geq 0$, put $\mathcal{R}(r) = \int_c^d \langle\rho\rangle_\theta\,\mathrm{d}z$. The compact positive-stress construction must satisfy

$$-\frac{15971}{122900}\,\mathcal{R}(r) \leq \mathfrak{c}(r) \leq \frac{19171}{122900}\,\mathcal{R}(r). \tag{71}$$

Both constants are attained by pointwise exact rational seven-ray allocations. The two-sided interval is asymmetric*; replacing it by a symmetric absolute-correction budget would impose a different condition.*

Proof. Average (65) over a complete angular period. The pressure term integrates to zero by periodicity, but $[\langle B\rangle_\theta]_c^d$ need not vanish. The radial identity follows by writing $\partial_r C + 2C/r = r^{-2}\partial_r(r^2C)$. The fundamental theorem of calculus gives (70). For the one-sided ray cone, the original seven coefficient floors imply the pointwise sharp interval

$$-\frac{15971}{122900}\rho \leq C \leq \frac{19171}{122900}\rho.$$

Its endpoints follow directly by combining the second, third, fourth and fifth ray inequalities; rational endpoint allocations show attainability. Integrating the pointwise interval proves (71). □

The torque-flux test is independent of the axial meridional test: it controls a different stress coordinate and has a different radial weight (r^2 in place of r). It applies directly only when the source frame is the standard cylindrical frame. For any additional spatial tilt, the covariant Cartesian source identity (63) and the rotated cone support must be used instead. Without authenticated H8 functions, neither flux condition is a completed source-matching test.

An independently checked Cartesian rotation example and exact two-cell rational support tests accompany these formulas. Neither a bound on δ_j nor (60) controls ∇E, so the weighted symmetric-divergence inverse and its derivative budget remain open.

The form of a compact cutoff may be fixed in a *test family* without identifying it with the source's unfrozen choice. Define

$$h(s) = \begin{cases} 0, & s \leq 0, \\ \dfrac{\exp(-1/s)}{\exp(-1/s) + \exp(-1/(1-s))}, & 0 < s < 1, \\ 1, & s \geq 1, \end{cases} \qquad \chi_*(r) = h(r-1)h(4-r).$$

Then χ_* is smooth and nonanalytic, zero outside $[1,4]$, identically one on $[2,3]$, and equals $1/2$ at $r = 3/2, 7/2$. As a radial function on the surrounding chart it is C_4-invariant; the hole at $r \leq 1$ makes its extension across the axis smooth. This provides a reproducible candidate for testing derivative estimates, not a determination of the H8 similarity-coordinate cutoff.

The source's toroidal pressure prescription likewise fixes only

$$\partial_r p_T = \frac{2u_\theta T_\theta + T_\theta^2}{r}. \tag{72}$$

For a chosen inner-radius normalization $p_T(a,z)=0$, its representative is

$$p_T(r,z)=\int_a^r \frac{2u_\theta(s,z)T_\theta(s,z)+T_\theta(s,z)^2}{s}\,\mathrm{d}s.$$

For $a=1$, $u_\theta=r$ and $T_\theta=rz(1-z)$ this gives

$$p_T=\frac{r^2-1}{2}\big(2z(1-z)+z^2(1-z)^2\big),\qquad [p_T]_{z=0}^{z=1}=0.$$

By contrast $T_\theta=rz$ gives the nonzero pressure jump $[p_T]_{-1}^{1}=2(r^2-1)$. The arbitrary radial integration trace $C(z)$ must be fixed by the remaining physical pressure equations; it is *not* an arbitrary gauge of the complete Navier–Stokes system.

To test all pieces in one local example, take $0\le z\le 1$, $U_r=-r$, $U_z=2z$, $U_\theta=r$, and the one-shaper data $K=a=1$, $\Xi=z^2(1-z)^2$, $b=(r-1)^3$, $\nu=1$, with the toroidal profile and pressure from the preceding zero-jump example. The C_4 core shear is partitioned into a diagonal $-r/4$ and ordered overlaps $-3r/4$; the shaper and viscous parts follow from Proposition 5.11. Directly integrating $K_{rz}=u_ru_z-(\partial_zu_r+\partial_ru_z)$ gives exactly

$$S(r)=\frac{-r^4-30r^3-2r+3}{30r^2},\qquad q_{\mathrm{s}}(r)=4,\qquad J_f(r)=\frac{-r^4+30r^3-32r+3}{30r^2}. \tag{73}$$

At $r=3/2$ the exact values are $S=-63/40$ and $J_f=91/120$. The independently computed axial source integral $F=(rS)'/r+q_{\mathrm{s}}$ reproduces J_f by integration from $a=1$. The example uses a *polynomial local core and shaper*, not χ_*; it verifies the algebraic combination and avoidance of double-counted core terms. The smooth nonanalytic cutoff and the authentic Case 006 transition, frame, and reference-centre fields still require their own complete weighted calculation.

6 The physical residual and terminal speed

Define the viscosity-one residual

$$\mathcal{N}(u,p)=u_t+(u\cdot\nabla)u-\Delta u+\nabla p. \tag{74}$$

For smooth (u,p) and (w,q), direct expansion yields

$$\mathcal{N}(u+w,p+q)=\mathcal{N}(u,p)+\mathcal{N}(w,q)+(u\cdot\nabla)w+(w\cdot\nabla)u. \tag{75}$$

Both nonlinear cross terms are essential. Equation (75), with explicit differentiability assumptions in the upstream physical definitions, is checked in Lean by the declaration `UnforcedPatching.navierStokesResidual_add`.

An exact positive symmetric-divergence inverse does not, by itself, cancel the unforced residual. For example, on the unit three-torus let $u=(\sin 2\pi y,0,0)$ and $p=0$. Then $\operatorname{div}u=0$ and $(u\cdot\nabla)u=0$, but

$$\mathcal{N}(u,0)=4\pi^2u\neq 0.$$

The symmetric periodic tensor

$$S=10I-2\pi\cos(2\pi y)(e_1\otimes e_2+e_2\otimes e_1)$$

is positive definite and satisfies $\operatorname{div}S=4\pi^2u$. This example concerns positive tensors in general, not the particular H8 seven-ray cone. It shows why a source-matched stress is only a precursor to a nonlinear correction: an actual divergence-free w and pressure q must solve

$$\partial_tw-\Delta w+(u\cdot\nabla)w+(w\cdot\nabla)u+(w\cdot\nabla)w+\nabla q=-\mathcal{N}(u,p)$$

with the necessary initial trace, periodic compatibility, regularity and terminal estimates.

Proposition 6.1 (Conditional terminal retention)**.** *Suppose that u has arbitrarily large speed at points with $t \uparrow 1$. If a correction w satisfies*

$$\|w(t,x)\| \le \alpha \|u(t,x)\| + C_0, \qquad 0 \le \alpha < 1, \quad C_0 \ge 0, \tag{76}$$

uniformly on $0 < t < 1$ and $x \in \mathbb{T}^3$, then $u + w$ also has arbitrarily large speed arbitrarily close to time one.

Proof. The reverse triangle inequality and (76) give

$$\|u + w\| \ge \|u\| - \|w\| \ge (1 - \alpha)\|u\| - C_0.$$

Given any target speed $M > 0$ and any terminal neighbourhood, choose a point therein where $\|u\| > (M + C_0)/(1 - \alpha)$. Then $\|u + w\| > M$ at that point. □

This last implication is kernel-checked by the Lean theorem

`UnforcedPatching.speedUnbounded_add_subleading`

It is deliberately conditional: no field u meeting the singularity premise, and no actual PDE correction w meeting (76), has been supplied by these lemmas.

7 What the formal development proves

The Lean project is pinned to Lean 4.34.0-rc2 and a fixed Mathlib revision. The checked physical interface states smoothness, three coordinate periods, zero forcing, nonzero initial data, finite fundamental-cell kinetic energy and endpoint speed divergence. Its existential proposition is named `UnforcedPatching.unforcedTargetStatement`; there is no proof term inhabiting that proposition.

The precise correspondence between the paper and the compiled `UnforcedPatching` declarations is as follows. In the second column, "missing" refers to mathematical constructions, not to unresolved Lean goals inside the named theorems.

Lean module	Checked scope and the missing construction
`Definitions`	Physical candidate predicates; no candidate witness.
`Residual`, `Correction`	Full residual addition and conditional cancellation; no constructed correcting field.
`Periodicity`, `PressureGauge`	Sum and time-only gauge identities; no periodic singular solution.
`ConeBudget`, `ConeOperator`, `SevenRayCenter`	Exact rational centre and sharp factor 14; no source-matched weighted positive stress.
`FourFieldPhysical`	Exact nine-entry ray-to-cylindrical tensor map modulo an isotropic gauge; entrywise four-coordinate bound in the frozen frame, not a spatially rotated PDE bridge.
`ExactCoefficientCone`, `SharpCoefficientDual`	Exact absolute-budget seven-coefficient polytope and sharp dual upper estimate; no frozen final Case 006 residual or spatial inverse.
`PositiveCoefficientCone`	Universal one-sided seven-ray positive-cone support bound; not a spatial stress inverse or a continuum integral formalization.

Lean module	Checked scope and the missing construction (continued)
`AxialFluxCone`, `JointFluxCone`	The separate axial/azimuthal caps and the coupled hexagon necessary bounds are finite algebra; new branch declarations require a fresh head-commit compilation and axiom receipt. The constructive hexagon lift is proved in the text and rational tests, not yet a continuum inverse.
`RotatingFrameDual`	Exact source-to-physical matrix Frobenius pullback and pointwise rotated one-sided dual bound; no derivative-of-frame or PDE source matching theorem.
`FrameConnectionDerivative`	Genuine entrywise product-rule derivative of EHE^{T}, skew angular connection and its commutator, including an exact moving-trigonometric-frame sign test; authentic H8 bounds and PDE source matching remain open.
`CompactGauge`	One-variable gauge derivative; no compact tensor support theorem.
`MeridionalInverse`	Actual axial and radial interval-integral FTC identities, geometric hoop cancellation, and pointwise radial/axial cylindrical inverse; no compact support theorem.
`Case006SourceBand`	Sharp finite-dimensional source-frame axial interval with an explicit one-sided-cone counterexample; no authentic spatial source identity.
`PressureCompensatedFlux`	Continuum radial FTC for the pressure-compensated axial source, conditional on the shear differential identity; axial pressure/tensor trace retained.
`CartesianAxialBridge`	Actual Cartesian axial divergence for axisymmetric mixed stress, including geometric K_{rz}/r and nonzero swirl; not a general tensor inverse.
`NonaxisymmetricAxialBridge`	Cartesian axial divergence with both radial and angular directional derivatives for arbitrary locally differentiable mixed stress; axial pressure derivative retained.
`AngularMean`	Exact periodic angular FTC removing the angular derivative after averaging, under explicit differentiability and integrability assumptions.
`PhysicalCircleAngular`	Physical circle chain rule and exact $-y\partial_x + x\partial_y$ correspondence, automatic angular endpoint equality, and integrated cancellation for Frechet-differentiable stress with integrable angular derivative.
`Retention`	Subleading corrections preserve terminal speed; neither required field constructed.

In particular, the residual-addition, correction-closure, seven-ray-floor, and terminal-retention declarations are all implications with explicit hypotheses. Their conjunction does not yield an inhabitant of `unforcedTargetStatement`. The printed axioms for the selected smoke-tested declarations are the standard logical axioms `propext`, `Classical.choice`, and `Quot.sound`.

In particular, `meridional_correction_divergence` checks both components of (3)–(4) for the hoop stress (9) and the actual interval-integral axial primitive (8), under stated local continuity and integrability assumptions. The separate declarations `radialMoment_relation` and `mixedCarrier_radial_balance` verify the antiderivative identity in (6) and its product carrier (7). The further theorem `sourceMatched_axialStress` checks the pointwise cylindrical source match $S_{zz} = \rho$, $f_z = \partial_z \rho$ used in Proposition 5.1. These results compile without proof admissions; selected axiom reports list only Lean's standard logical axioms. The further

`CartesianAxialBridge` and `NonaxisymmetricAxialBridge` theorems now establish the actual Cartesian *axial* divergence component, including the nonzero angular directional derivative. The `PhysicalCircleAngular` theorem derives that derivative by the Frechet chain rule along a genuine Cartesian circle, supplies the periodic endpoint equality, and invokes `AngularMean` to remove its contribution only *after* integration. The `FrameConnectionDerivative` theorem separately derives the full product-rule differentiation of rotated stresses, including the $\Omega H - H\Omega$ connection term; it does not assume that E is spatially constant. The `PressureCompensatedFlux` theorem proves the separate radial FTC with the full axial boundary trace retained. Derivation of the integrated source–shear differential hypothesis from authenticated H8 profiles, all necessary interchange-of-differentiation conditions, authentic H8 frame profiles and uniform connection estimates, the complete Cartesian tensor divergence, global support, moment compatibility and weighted positive inverse remain open. Thus the full restricted Theorem 2.1 is *partially*, not fully, formalized.

An independent exact-arithmetic test evaluates a nontrivial annular polynomial instance of the radial compatibility identity. For the uncompensated purely radial version, the curl obstruction at $(r, z) = (5/4, 1/2)$ equals $-5103/32768$ exactly; the test explicitly rejects the pressure-neutral interpretation in this case. The polynomial is used only for interior differentiation, not as an infinitely smooth zero-extended cutoff.

A second independent exact-rational test reproduces (21)–(24) from polynomial integrals and checks the complete cylindrical divergence of the dual test field. An additional exact reconstruction and Lean theorem prove the fixed-frame four-field map (16), its inverse coordinate recovery and its entrywise consequence; the test (25) records the failure of this *conservative gradient method* at the original coefficient budget. The sharper polytope method proves Proposition 5.4: a 40×40 Bernstein partition yields the exact rational upper bound (28). No executable diagnostic substitutes for frozen final Case 006 profiles and rotated-frame PDE estimates. The recovered 122-page revision permits an additional standard-library rational check of Proposition 5.10: it verifies both interval endpoints, an attaining absolute-budget tensor, a one-sided-positive witness exceeding that budget, all seven coefficient floors, and a nonzero pressure boundary primitive. These are finite algebra checks, not an authentic-source inequality evaluation. The equivalence (17) is independently checked on exact rational samples and kernel-checked by `original_seven_coefficient_cone_exact`. The additional kernel theorem `original_cone_dual_upper` establishes the finite-dimensional upper bound in (26); exact rational maximizing witnesses independently verify sharpness. The Bernstein integral itself is a rational Python certificate, not a Lean theorem.

The separate one-sided cone is described by (29); its universal support upper bound is proved by the compiled Lean declaration `one_sided_positive_cone_dual_upper`. Independent exact rational feasible witnesses attain this value. A new 40×40 rational partition proves the lower bound (34), with a separate signed Bernstein upper enclosure. The integral certificate and its comparison with the source pairing are not formalized in Lean. The strict positivity of $J - L$ means only that this *particular* dual test supplies no one-sided positive-cone nonexistence certificate.

The further compiled declarations

`rotating_fourField_frobenius`
`rotated_oneSided_dual_upper`

establish (36) and (37) for every pointwise 3×3 frame matrix. Independent rational tests verify the rotated covector, equality of matrix contractions, an attainable one-sided dual support, and preservation of isotropic pressure gauges under orthogonal (but not arbitrary) frames. The spatial differential identity (39) and integrated inequality (38) are proved on paper, not yet in Lean; neither has been evaluated against the still-unfrozen complete Case 006 prepared stress.

The source code, machine-readable G0–G8 ledger, smoke tests and reproduction commands accompany the public integration branch [6]. A prior local pinned build passed sixteen smoke modules, and the main physical Lean workflow later received a GitHub-hosted success receipt. A separate weighted source-flux workflow halted during Lean toolchain installation before executing any mathematical check. None of these receipts covers the new rational source-band test until the updated branch is independently checked. Compilation does not replace semantic theorem auditing or independent scrutiny of the paper proof.

8 Remaining analytic requirements

The compact meridional inverse of Theorem 2.1 is a properly restricted solvability result. It does not resolve the weighted attachment problem because the desired source may have an azimuthal component, a noncompact Leray-projected tail, nontrivial pressure coupling, or incompatible moments. Even when compact inversion is possible, the output may leave the required seven-ray positive cone, especially as the support shrinks. The strict dual obstruction in Proposition 5.1, together with Corollary 5.2, excludes the sufficient four-coordinate estimate; Proposition 5.4 also excludes the original absolute coefficient correction budget for that same manufactured source. In contrast, Proposition 5.7 shows that the tested dual field has no obstruction for the larger one-sided cone; this leaves other dual tests and all primal construction questions open. The next mathematical question is whether the *authentic* source, perhaps using a tilted frame or a more flexible coefficient allocation, admits a weighted inverse satisfying the one-sided positive H8 ray conditions (or the absolute inequalities (17)) and all spatial derivative bounds. The sharp cone description enables a direct weighted dual optimization against the authentic source through (38), without replacing those inequalities by the factor-14 sufficient estimate. No numerical conclusion can be drawn from this inequality until the actual g, ρ, R_*, E are supplied and the frame-derivative terms in (39) are controlled.

The joint constraints (31) supply a sharper source-facing test in an aligned cylindrical frame. In a moving physical frame they must be replaced by the spatially integrated support of the complete rotated covector; the corrected Proposition 5.16 makes the needed angular test variation explicit. Failure of a fully certified necessary inequality would obstruct the proposed source-matched positive inverse. Passing these necessary tests would not construct that inverse and would not discharge the physical nonlinear correction.

A complete unforced singularity construction must still provide a single authenticated local profile, source-matched smooth shell data, a scale-sharp weighted symmetric inverse respecting the coefficient margin, a periodic nonlinear correction with a proved right-inverse/contraction scheme, the full physical zero-residual identity, and terminal retention for that actual correction. Each claim requires a further source-to-Lean semantic audit and a compiled theorem using the genuine PDE definitions.

Thus the present mathematical result is an exact compact stress construction and several independently checked necessary or conditional ingredients. The existence or nonexistence of a finite-time singular solution to (1) with smooth nonzero periodic initial data is *not decided by this manuscript.*

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